James (Cheng-Wen) Fu

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Computational Design — *Aug. 2025 - May. 2027*

National Cheng Kung University — **Tainan, Taiwan**
Bachelor of Architectural Design — *Sep. 2018 - Jun. 2023*

WORK EXPERIENCE

Design Technology Intern | Gensler — **New York, NY**
Architects & Planners Firm, Intern — *Jun. 2026 - Aug. 2026*

- Built an NLP pipeline to classify 7,390 Design Technology support tickets, including Revit-related issues, achieving 86.6% accuracy and 0.806 Macro F1 to improve ticket categorization and analysis.
- Contributed to a building benchmarking project by developing Dynamo workflows to clean, label, and standardize Revit room and area data across 20+ projects for visualization and analysis.

Modeler and Computational Designer | Illoca — **Remote**
AI Startup in the AEC Industry, Contractor — *Apr. 2024 - Aug. 2025*

- Created large-scale Revit models to develop structured 3D training datasets for AI-driven architectural design.
- Built parametric façade systems in Grasshopper to generate diverse design variations for AI model training.

Research Assistant | National Taiwan University — **Taipei, Taiwan**
CAE Division, Department of Civil Engineering — *Dec. 2023 - Jul. 2025*

- Engineered a full-stack smart campus platform using React, Node.js, and PostgreSQL, connecting BIM model elements with energy data through REST APIs for interactive 3D visualization and building performance analysis.
- Developed Razor-based interfaces integrating Autodesk Viewer with a SQL-backed materials database, enabling material search, filtering, and spatial visualization within revit models.

Computational Designer | J. J. Pan and Partners — **Taipei, Taiwan**
Architects & Planners Firm, Part-time — *Apr. 2024 - Apr. 2025*

- Developed a Grasshopper plugin with green building experts, integrating material-based carbon calculations, interactive 3D visualization, and data dashboards for real-time LEBR assessments across two client projects.
- Co-edited the firm's annual publication and contributed to its future AI strategy for architectural workflows.

PUBLICATION

Physical AI: VLM-based Planning and Behavioral Evaluation
Co-Author, ACM CHI Conference on Human Factors in Computing Systems (CHI) 2027, Under Review

- Built an AI agent integrating YOLO, SAM2, and VLM-based planning for environment-aware interactions.
- Conducted a participant study evaluating human perceptions of agent behavior and environmental awareness.

LLM-Driven Agentic Tool Creation for Visual Programming
First Author, Association for Computer Aided Design in Architecture (ACADIA) 2026, Fieldnote

- Built an MCP bridge, agent-callable tools, and custom skills to connect LLM agents with Grasshopper.
- Proposed an agentic workflow shifting visual programming from node assembly to AI-driven tool creation and reuse.

Streamlining Digital Twin Deployment and Data Management Platform
First Author, International Conference on Civil and Building Engineering Informatics (ICCBEI) 2025, Full Paper

- Built a web platform using APS APIs for Revit integration, cloud revit model management, and data visualization.
- Designed a role-based workflow coordinating user permissions, data contributions, and digital twin management.

SKILLS

Programming & Web: Python, JavaScript, React, Node.js, REST APIs, PostgreSQL

AI & Machine Learning: PyTorch, Model Training & Evaluation, LLM Agents, MCP

Automation & Visualization: Grasshopper, Dynamo, Revit, Autodesk APS, BIM Visualization, Figma